

## 3.1 Cohomology Groups

Homology groups  $H_n(X)$  are the result of a two-stage process: First one forms a chain complex  $\cdots \rightarrow C_n \xrightarrow{\partial} C_{n-1} \rightarrow \cdots$  of singular, simplicial, or cellular chains, then one takes the homology groups of this chain complex,  $\text{Ker } \partial / \text{Im } \partial$ . To obtain the cohomology groups  $H^n(X; G)$  we interpolate an intermediate step, replacing the chain groups  $C_n$  by the dual groups  $\text{Hom}(C_n, G)$  and the boundary maps  $\partial$  by their dual maps  $\delta$ , before forming the cohomology groups  $\text{Ker } \delta / \text{Im } \delta$ . The plan for this section is first to sort out the algebra of this dualization process and show that the cohomology groups are determined algebraically by the homology groups, though in a somewhat subtle way. Then after this algebraic excursion we will define the cohomology groups of spaces and show that these satisfy basic properties very much like those for homology. The payoff for all this formal work will begin to be apparent in subsequent sections.

### The Universal Coefficient Theorem

Let us begin with a simple example. Consider the chain complex

$$0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0$$

$$\quad \quad \quad \parallel \quad \quad \parallel \quad \quad \parallel \quad \quad \parallel$$

$$\quad \quad \quad C_3 \quad \quad C_2 \quad \quad C_1 \quad \quad C_0$$

where  $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$  is the map  $x \mapsto 2x$ . If we dualize by taking  $\text{Hom}(-, G)$  with  $G = \mathbb{Z}$ , we obtain the cochain complex

$$0 \longleftarrow \mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{2} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \longleftarrow 0$$

$$\quad \quad \quad \parallel \quad \quad \parallel \quad \quad \parallel \quad \quad \parallel$$

$$\quad \quad \quad C_3^* \quad \quad C_2^* \quad \quad C_1^* \quad \quad C_0^*$$

In the original chain complex the homology groups are  $\mathbb{Z}$ 's in dimensions 0 and 3, together with a  $\mathbb{Z}_2$  in dimension 1. The homology groups of the dual cochain complex, which are called cohomology groups to emphasize the dualization, are again  $\mathbb{Z}$ 's in dimensions 0 and 3, but the  $\mathbb{Z}_2$  in the 1-dimensional homology of the original complex has shifted up a dimension to become a  $\mathbb{Z}_2$  in 2-dimensional cohomology.

More generally, consider any chain complex of finitely generated free abelian groups. Such a chain complex always splits as the direct sum of elementary complexes of the forms  $0 \rightarrow \mathbb{Z} \rightarrow 0$  and  $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow 0$ , according to Exercise 43 in §2.2. Applying  $\text{Hom}(-, \mathbb{Z})$  to this direct sum of elementary complexes, we obtain the direct sum of the corresponding dual complexes  $0 \leftarrow \mathbb{Z} \leftarrow 0$  and  $0 \leftarrow \mathbb{Z} \xleftarrow{m} \mathbb{Z} \leftarrow 0$ . Thus the cohomology groups are the same as the homology groups except that torsion is shifted up one dimension. We will see later in this section that the same relation between homology and cohomology holds whenever the homology groups are finitely generated, even when the chain groups are not finitely generated. It would also be quite easy to

see in this example what happens if  $\text{Hom}(-, \mathbb{Z})$  is replaced by  $\text{Hom}(-, G)$ , since the dual elementary cochain complexes would then be  $0 \leftarrow G \leftarrow 0$  and  $0 \leftarrow G \xleftarrow{m} G \leftarrow 0$ .

Consider now a completely general chain complex  $C$  of free abelian groups

$$\cdots \longrightarrow C_{n+1} \xrightarrow{\partial} C_n \xrightarrow{\partial} C_{n-1} \longrightarrow \cdots$$

To dualize this complex we replace each chain group  $C_n$  by its dual **cochain group**  $C_n^* = \text{Hom}(C_n, G)$ , the group of homomorphisms  $C_n \rightarrow G$ , and we replace each boundary map  $\partial: C_n \rightarrow C_{n-1}$  by its dual **coboundary map**  $\delta = \partial^*: C_{n-1}^* \rightarrow C_n^*$ . The reason why  $\delta$  goes in the opposite direction from  $\partial$ , increasing rather than decreasing dimension, is purely formal: For a homomorphism  $\alpha: A \rightarrow B$ , the dual homomorphism  $\alpha^*: \text{Hom}(B, G) \rightarrow \text{Hom}(A, G)$  is defined by  $\alpha^*(\varphi) = \varphi\alpha$ , so  $\alpha^*$  sends  $B \xrightarrow{\varphi} G$  to the composition  $A \xrightarrow{\alpha} B \xrightarrow{\varphi} G$ . Dual homomorphisms obviously satisfy  $(\alpha\beta)^* = \beta^*\alpha^*$ ,  $1^* = 1$ , and  $0^* = 0$ . In particular, since  $\partial\partial = 0$  it follows that  $\delta\delta = 0$ , and the **cohomology group**  $H^n(C; G)$  can be defined as the ‘homology group’  $\text{Ker } \delta / \text{Im } \delta$  at  $C_n^*$  in the cochain complex

$$\cdots \longleftarrow C_{n+1}^* \xleftarrow{\delta} C_n^* \xleftarrow{\delta} C_{n-1}^* \longleftarrow \cdots$$

Our goal is to show that the cohomology groups  $H^n(C; G)$  are determined solely by  $G$  and the homology groups  $H_n(C) = \text{Ker } \partial / \text{Im } \partial$ . A first guess might be that  $H^n(C; G)$  is isomorphic to  $\text{Hom}(H_n(C), G)$ , but this is overly optimistic, as shown by the example above where  $H_2$  was zero while  $H^2$  was nonzero. Nevertheless, there is a natural map  $h: H^n(C; G) \rightarrow \text{Hom}(H_n(C), G)$ , defined as follows. Denote the cycles and boundaries by  $Z_n = \text{Ker } \partial \subset C_n$  and  $B_n = \text{Im } \partial \subset C_n$ . A class in  $H^n(C; G)$  is represented by a homomorphism  $\varphi: C_n \rightarrow G$  such that  $\delta\varphi = 0$ , that is,  $\varphi\partial = 0$ , or in other words,  $\varphi$  vanishes on  $B_n$ . The restriction  $\varphi_0 = \varphi|_{Z_n}$  then induces a quotient homomorphism  $\overline{\varphi}_0: Z_n/B_n \rightarrow G$ , an element of  $\text{Hom}(H_n(C), G)$ . If  $\varphi$  is in  $\text{Im } \delta$ , say  $\varphi = \delta\psi = \psi\partial$ , then  $\varphi$  is zero on  $Z_n$ , so  $\varphi_0 = 0$  and hence also  $\overline{\varphi}_0 = 0$ . Thus there is a well-defined quotient map  $h: H^n(C; G) \rightarrow \text{Hom}(H_n(C), G)$  sending the cohomology class of  $\varphi$  to  $\overline{\varphi}_0$ . Obviously  $h$  is a homomorphism.

It is not hard to see that  $h$  is surjective. The short exact sequence

$$0 \longrightarrow Z_n \longrightarrow C_n \xrightarrow{\partial} B_{n-1} \longrightarrow 0$$

splits since  $B_{n-1}$  is free, being a subgroup of the free abelian group  $C_{n-1}$ . Thus there is a projection homomorphism  $p: C_n \rightarrow Z_n$  that restricts to the identity on  $Z_n$ . Composing with  $p$  gives a way of extending homomorphisms  $\varphi_0: Z_n \rightarrow G$  to homomorphisms  $\varphi = \varphi_0 p: C_n \rightarrow G$ . In particular, this extends homomorphisms  $Z_n \rightarrow G$  that vanish on  $B_n$  to homomorphisms  $C_n \rightarrow G$  that still vanish on  $B_n$ , or in other words, it extends homomorphisms  $H_n(C) \rightarrow G$  to elements of  $\text{Ker } \delta$ . Thus we have a homomorphism  $\text{Hom}(H_n(C), G) \rightarrow \text{Ker } \delta$ . Composing this with the quotient map  $\text{Ker } \delta \rightarrow H^n(C; G)$  gives a homomorphism from  $\text{Hom}(H_n(C), G)$  to  $H^n(C; G)$ . If we

follow this map by  $h$  we get the identity map on  $\text{Hom}(H_n(C), G)$  since the effect of composing with  $h$  is simply to undo the effect of extending homomorphisms via  $p$ . This shows that  $h$  is surjective. In fact it shows that we have a split short exact sequence

$$0 \rightarrow \text{Ker } h \rightarrow H^n(C; G) \xrightarrow{h} \text{Hom}(H_n(C), G) \rightarrow 0$$

The remaining task is to analyze  $\text{Ker } h$ . A convenient way to start the process is to consider not just the chain complex  $C$ , but also its subcomplexes consisting of the cycles and the boundaries. Thus we consider the commutative diagram of short exact sequences

$$(i) \quad \begin{array}{ccccccc} 0 & \longrightarrow & Z_{n+1} & \longrightarrow & C_{n+1} & \xrightarrow{\partial} & B_n \longrightarrow 0 \\ & & \downarrow 0 & & \downarrow \partial & & \downarrow 0 \\ 0 & \longrightarrow & Z_n & \longrightarrow & C_n & \xrightarrow{\partial} & B_{n-1} \longrightarrow 0 \end{array}$$

where the vertical boundary maps on  $Z_{n+1}$  and  $B_n$  are the restrictions of the boundary map in the complex  $C$ , hence are zero. Dualizing (i) gives a commutative diagram

$$(ii) \quad \begin{array}{ccccccc} 0 & \longleftarrow & Z_{n+1}^* & \longleftarrow & C_{n+1}^* & \longleftarrow & B_n^* \longleftarrow 0 \\ & & \uparrow 0 & & \uparrow \delta & & \uparrow 0 \\ 0 & \longleftarrow & Z_n^* & \longleftarrow & C_n^* & \longleftarrow & B_{n-1}^* \longleftarrow 0 \end{array}$$

The rows here are exact since, as we have already remarked, the rows of (i) split, and the dual of a split short exact sequence is a split short exact sequence because of the natural isomorphism  $\text{Hom}(A \oplus B, G) \approx \text{Hom}(A, G) \oplus \text{Hom}(B, G)$ .

We may view (ii), like (i), as part of a short exact sequence of chain complexes. Since the coboundary maps in the  $Z_n^*$  and  $B_n^*$  complexes are zero, the associated long exact sequence of homology groups has the form

$$(iii) \quad \cdots \longleftarrow B_n^* \longleftarrow Z_n^* \longleftarrow H^n(C; G) \longleftarrow B_{n-1}^* \longleftarrow Z_{n-1}^* \longleftarrow \cdots$$

The 'boundary maps'  $Z_n^* \rightarrow B_n^*$  in this long exact sequence are in fact the dual maps  $i_n^*$  of the inclusions  $i_n: B_n \rightarrow Z_n$ , as one sees by recalling how these boundary maps are defined: In (ii) one takes an element of  $Z_n^*$ , pulls this back to  $C_n^*$ , applies  $\delta$  to get an element of  $C_{n+1}^*$ , then pulls this back to  $B_n^*$ . The first of these steps extends a homomorphism  $\varphi_0: Z_n \rightarrow G$  to  $\varphi: C_n \rightarrow G$ , the second step composes this  $\varphi$  with  $\partial$ , and the third step undoes this composition and restricts  $\varphi$  to  $B_n$ . The net effect is just to restrict  $\varphi_0$  from  $Z_n$  to  $B_n$ .

A long exact sequence can always be broken up into short exact sequences, and doing this for the sequence (iii) yields short exact sequences

$$(iv) \quad 0 \longleftarrow \text{Ker } i_n^* \longleftarrow H^n(C; G) \longleftarrow \text{Coker } i_{n-1}^* \longleftarrow 0$$

The group  $\text{Ker } i_n^*$  can be identified naturally with  $\text{Hom}(H_n(C), G)$  since elements of  $\text{Ker } i_n^*$  are homomorphisms  $Z_n \rightarrow G$  that vanish on the subgroup  $B_n$ , and such homomorphisms are the same as homomorphisms  $Z_n/B_n \rightarrow G$ . Under this identification of

$\text{Ker } i_n^*$  with  $\text{Hom}(H_n(C), G)$ , the map  $H^n(C; G) \rightarrow \text{Ker } i_n^*$  in (iv) becomes the map  $h$  considered earlier. Thus we can rewrite (iv) as a split short exact sequence

$$(v) \quad 0 \rightarrow \text{Coker } i_{n-1}^* \rightarrow H^n(C; G) \xrightarrow{h} \text{Hom}(H_n(C), G) \rightarrow 0$$

Our objective now is to show that the more mysterious term  $\text{Coker } i_{n-1}^*$  depends only on  $H_{n-1}(C)$  and  $G$ , in a natural, functorial way. First let us observe that  $\text{Coker } i_{n-1}^*$  would be zero if it were always true that the dual of a short exact sequence was exact, since the dual of the short exact sequence

$$(vi) \quad 0 \rightarrow B_{n-1} \xrightarrow{i_{n-1}} Z_{n-1} \rightarrow H_{n-1}(C) \rightarrow 0$$

is the sequence

$$(vii) \quad 0 \leftarrow B_{n-1}^* \xleftarrow{i_{n-1}^*} Z_{n-1}^* \leftarrow H_{n-1}(C)^* \leftarrow 0$$

and if this were exact at  $B_{n-1}^*$ , then  $i_{n-1}^*$  would be surjective, hence  $\text{Coker } i_{n-1}^*$  would be zero. This argument does apply if  $H_{n-1}(C)$  happens to be free, since (vi) splits in this case, which implies that (vii) is also split exact. So in this case the map  $h$  in (v) is an isomorphism. However, in the general case it is easy to find short exact sequences whose duals are not exact. For example, if we dualize  $0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}_n \rightarrow 0$  by applying  $\text{Hom}(-, \mathbb{Z})$  we get  $0 \leftarrow \mathbb{Z} \xleftarrow{n} \mathbb{Z} \leftarrow 0 \leftarrow 0$  which fails to be exact at the left-hand  $\mathbb{Z}$ , precisely the place we are interested in for  $\text{Coker } i_{n-1}^*$ .

We might mention in passing that the loss of exactness at the left end of a short exact sequence after dualization is in fact all that goes wrong, in view of the following:

**Exercise.** If  $A \rightarrow B \rightarrow C \rightarrow 0$  is exact, then dualizing by applying  $\text{Hom}(-, G)$  yields an exact sequence  $A^* \leftarrow B^* \leftarrow C^* \leftarrow 0$ .

However, we will not need this fact in what follows.

The exact sequence (vi) has the special feature that both  $B_{n-1}$  and  $Z_{n-1}$  are free, so (vi) can be regarded as a free resolution of  $H_{n-1}(C)$ , where a **free resolution** of an abelian group  $H$  is an exact sequence

$$\cdots \rightarrow F_2 \xrightarrow{f_2} F_1 \xrightarrow{f_1} F_0 \xrightarrow{f_0} H \rightarrow 0$$

with each  $F_n$  free. If we dualize this free resolution by applying  $\text{Hom}(-, G)$ , we may lose exactness, but at least we get a chain complex — or perhaps we should say ‘cochain complex,’ but algebraically there is no difference. This dual complex has the form

$$\cdots \leftarrow F_2^* \xleftarrow{f_2^*} F_1^* \xleftarrow{f_1^*} F_0^* \xleftarrow{f_0^*} H^* \leftarrow 0$$

Let us use the temporary notation  $H^n(F; G)$  for the homology group  $\text{Ker } f_{n+1}^* / \text{Im } f_n^*$  of this dual complex. Note that the group  $\text{Coker } i_{n-1}^*$  that we are interested in is  $H^1(F; G)$  where  $F$  is the free resolution in (vi). Part (b) of the following lemma therefore shows that  $\text{Coker } i_{n-1}^*$  depends only on  $H_{n-1}(C)$  and  $G$ .

**Lemma 3.1. (a)** Given free resolutions  $F$  and  $F'$  of abelian groups  $H$  and  $H'$ , then every homomorphism  $\alpha: H \rightarrow H'$  can be extended to a chain map from  $F$  to  $F'$ :

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & F_2 & \xrightarrow{f_2} & F_1 & \xrightarrow{f_1} & F_0 & \xrightarrow{f_0} & H & \longrightarrow & 0 \\ & & \downarrow \alpha_2 & & \downarrow \alpha_1 & & \downarrow \alpha_0 & & \downarrow \alpha & & \\ \cdots & \longrightarrow & F'_2 & \xrightarrow{f'_2} & F'_1 & \xrightarrow{f'_1} & F'_0 & \xrightarrow{f'_0} & H' & \longrightarrow & 0 \end{array}$$

Furthermore, any two such chain maps extending  $\alpha$  are chain homotopic.

**(b)** For any two free resolutions  $F$  and  $F'$  of  $H$ , there are canonical isomorphisms  $H^n(F; G) \approx H^n(F'; G)$  for all  $n$ .

**Proof:** The  $\alpha_i$ 's will be constructed inductively. Since the  $F_i$ 's are free, it suffices to define each  $\alpha_i$  on a basis for  $F_i$ . To define  $\alpha_0$ , observe that surjectivity of  $f'_0$  implies that for each basis element  $x$  of  $F_0$  there exists  $x' \in F'_0$  such that  $f'_0(x') = \alpha f_0(x)$ , so we define  $\alpha_0(x) = x'$ . We would like to define  $\alpha_1$  in the same way, sending a basis element  $x \in F_1$  to an element  $x' \in F'_1$  such that  $f'_1(x') = \alpha_0 f_1(x)$ . Such an  $x'$  will exist if  $\alpha_0 f_1(x)$  lies in  $\text{Im } f'_1 = \text{Ker } f'_0$ , which it does since  $f'_0 \alpha_0 f_1 = \alpha f_0 f_1 = 0$ . The same procedure defines all the subsequent  $\alpha_i$ 's.

If we have another chain map extending  $\alpha$  given by maps  $\alpha'_i: F_i \rightarrow F'_i$ , then the differences  $\beta_i = \alpha_i - \alpha'_i$  define a chain map extending the zero map  $\beta: H \rightarrow H'$ . It will suffice to construct maps  $\lambda_i: F_i \rightarrow F'_{i+1}$  defining a chain homotopy from  $\beta_i$  to 0, that is, with  $\beta_i = f'_{i+1} \lambda_i + \lambda_{i-1} f_i$ . The  $\lambda_i$ 's are constructed inductively by a procedure much like the construction of the  $\alpha_i$ 's. When  $i = 0$  we let  $\lambda_{-1}: H \rightarrow F'_0$  be zero, and then the desired relation becomes  $\beta_0 = f'_1 \lambda_0$ . We can achieve this by letting  $\lambda_0$  send a basis element  $x$  to an element  $x' \in F'_1$  such that  $f'_1(x') = \beta_0(x)$ . Such an  $x'$  exists since  $\text{Im } f'_1 = \text{Ker } f'_0$  and  $f'_0 \beta_0(x) = \beta f_0(x) = 0$ . For the inductive step we wish to define  $\lambda_i$  to take a basis element  $x \in F_i$  to an element  $x' \in F'_{i+1}$  such that  $f'_{i+1}(x') = \beta_i(x) - \lambda_{i-1} f_i(x)$ . This will be possible if  $\beta_i(x) - \lambda_{i-1} f_i(x)$  lies in  $\text{Im } f'_{i+1} = \text{Ker } f'_i$ , which will hold if  $f'_i(\beta_i - \lambda_{i-1} f_i) = 0$ . Using the relation  $f'_i \beta_i = \beta_{i-1} f_i$  and the relation  $\beta_{i-1} = f'_i \lambda_{i-1} + \lambda_{i-2} f_{i-1}$  which holds by induction, we have

$$\begin{aligned} f'_i(\beta_i - \lambda_{i-1} f_i) &= f'_i \beta_i - f'_i \lambda_{i-1} f_i \\ &= \beta_{i-1} f_i - f'_i \lambda_{i-1} f_i = (\beta_{i-1} - f'_i \lambda_{i-1}) f_i = \lambda_{i-2} f_{i-1} f_i = 0 \end{aligned}$$

as desired. This finishes the proof of (a).

The maps  $\alpha_n$  constructed in (a) dualize to maps  $\alpha_n^*: F_n^* \rightarrow F_n^*$  forming a chain map between the dual complexes  $F'^*$  and  $F^*$ . Therefore we have induced homomorphisms on cohomology  $\alpha^*: H^n(F'; G) \rightarrow H^n(F; G)$ . These do not depend on the choice of  $\alpha_n$ 's since any other choices  $\alpha'_n$  are chain homotopic, say via chain homotopies  $\lambda_n$ , and then  $\alpha_n^*$  and  $\alpha_n'^*$  are chain homotopic via the dual maps  $\lambda_n^*$  since the dual of the relation  $\alpha_i - \alpha'_i = f'_{i+1} \lambda_i + \lambda_{i-1} f_i$  is  $\alpha_i^* - \alpha_i'^* = \lambda_i^* f_{i+1}^* + f_i^* \lambda_{i-1}^*$ .

The induced homomorphisms  $\alpha^*: H^n(F'; G) \rightarrow H^n(F; G)$  satisfy  $(\beta \alpha)^* = \alpha^* \beta^*$  for a composition  $H \xrightarrow{\alpha} H' \xrightarrow{\beta} H''$  with a free resolution  $F''$  of  $H''$  also given, since

one can choose the compositions  $\beta_n \alpha_n$  of extensions  $\alpha_n$  of  $\alpha$  and  $\beta_n$  of  $\beta$  as an extension of  $\beta\alpha$ . In particular, if we take  $\alpha$  to be an isomorphism and  $\beta$  to be its inverse, with  $F'' = F$ , then  $\alpha^* \beta^* = (\beta\alpha)^* = \mathbb{1}$ , the latter equality coming from the obvious extension of  $\mathbb{1} : H \rightarrow H$  by the identity map of  $F$ . The same reasoning shows  $\beta^* \alpha^* = \mathbb{1}$ , so  $\alpha^*$  is an isomorphism. Finally, if we specialize further, taking  $\alpha$  to be the identity but with two different free resolutions  $F$  and  $F'$ , we get a canonical isomorphism  $\mathbb{1}^* : H^n(F'; G) \rightarrow H^n(F; G)$ .  $\square$

Every abelian group  $H$  has a free resolution of the form  $0 \rightarrow F_1 \rightarrow F_0 \rightarrow H \rightarrow 0$ , with  $F_i = 0$  for  $i > 1$ , obtainable in the following way. Choose a set of generators for  $H$  and let  $F_0$  be a free abelian group with basis in one-to-one correspondence with these generators. Then we have a surjective homomorphism  $f_0 : F_0 \rightarrow H$  sending the basis elements to the chosen generators. The kernel of  $f_0$  is free, being a subgroup of a free abelian group, so we can let  $F_1$  be this kernel with  $f_1 : F_1 \rightarrow F_0$  the inclusion, and we can then take  $F_i = 0$  for  $i > 1$ . For this free resolution we obviously have  $H^n(F; G) = 0$  for  $n > 1$ , so this must also be true for all free resolutions. Thus the only interesting group  $H^n(F; G)$  is  $H^1(F; G)$ . As we have seen, this group depends only on  $H$  and  $G$ , and the standard notation for it is  $\text{Ext}(H, G)$ . This notation arises from the fact that  $\text{Ext}(H, G)$  has an interpretation as the set of isomorphism classes of extensions of  $G$  by  $H$ , that is, short exact sequences  $0 \rightarrow G \rightarrow J \rightarrow H \rightarrow 0$ , with a natural definition of isomorphism between such exact sequences. This is explained in books on homological algebra, for example [Brown 1982], [Hilton & Stammach 1970], or [MacLane 1963]. However, this interpretation of  $\text{Ext}(H, G)$  is rarely needed in algebraic topology.

Summarizing, we have established the following algebraic result:

**Theorem 3.2.** *If a chain complex  $C$  of free abelian groups has homology groups  $H_n(C)$ , then the cohomology groups  $H^n(C; G)$  of the cochain complex  $\text{Hom}(C_n, G)$  are determined by split exact sequences*

$$0 \rightarrow \text{Ext}(H_{n-1}(C), G) \rightarrow H^n(C; G) \xrightarrow{h} \text{Hom}(H_n(C), G) \rightarrow 0 \quad \square$$

This is known as the **universal coefficient theorem for cohomology** because it is formally analogous to the universal coefficient theorem for homology in §3.A which expresses homology with arbitrary coefficients in terms of homology with  $\mathbb{Z}$  coefficients.

Computing  $\text{Ext}(H, G)$  for finitely generated  $H$  is not difficult using the following three properties:

- $\text{Ext}(H \oplus H', G) \approx \text{Ext}(H, G) \oplus \text{Ext}(H', G)$ .
- $\text{Ext}(H, G) = 0$  if  $H$  is free.
- $\text{Ext}(\mathbb{Z}_n, G) \approx G/nG$ .

The first of these can be obtained by using the direct sum of free resolutions of  $H$  and  $H'$  as a free resolution for  $H \oplus H'$ . If  $H$  is free, the free resolution  $0 \rightarrow H \rightarrow H \rightarrow 0$

yields the second property, while the third comes from dualizing the free resolution  $0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}_n \rightarrow 0$  to produce an exact sequence

$$\begin{array}{ccccccc} 0 & \leftarrow & \text{Ext}(\mathbb{Z}_n, G) & \leftarrow & \text{Hom}(\mathbb{Z}, G) & \xleftarrow{n} & \text{Hom}(\mathbb{Z}, G) \leftarrow \text{Hom}(\mathbb{Z}_n, G) \leftarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ & & G/nG & \leftarrow & G & \xleftarrow{n} & G \end{array}$$

In particular, these three properties imply that  $\text{Ext}(H, \mathbb{Z})$  is isomorphic to the torsion subgroup of  $H$  if  $H$  is finitely generated. Since  $\text{Hom}(H, \mathbb{Z})$  is isomorphic to the free part of  $H$  if  $H$  is finitely generated, we have:

**Corollary 3.3.** *If the homology groups  $H_n$  and  $H_{n-1}$  of a chain complex  $C$  of free abelian groups are finitely generated, with torsion subgroups  $T_n \subset H_n$  and  $T_{n-1} \subset H_{n-1}$ , then  $H^n(C; \mathbb{Z}) \approx (H_n/T_n) \oplus T_{n-1}$ .  $\square$*

It is useful in many situations to know that the short exact sequences in the universal coefficient theorem are natural, meaning that a chain map  $\alpha$  between chain complexes  $C$  and  $C'$  of free abelian groups induces a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Ext}(H_{n-1}(C), G) & \longrightarrow & H^n(C; G) & \xrightarrow{h} & \text{Hom}(H_n(C), G) \longrightarrow 0 \\ & & \uparrow (\alpha_*)^* & & \uparrow \alpha^* & & \uparrow (\alpha_*)^* \\ 0 & \longrightarrow & \text{Ext}(H_{n-1}(C'), G) & \longrightarrow & H^n(C'; G) & \xrightarrow{h} & \text{Hom}(H_n(C'), G) \longrightarrow 0 \end{array}$$

This is apparent if one just thinks about the construction; one obviously obtains a map between the short exact sequences (iv) containing  $\text{Ker } i_n^*$  and  $\text{Coker } i_{n-1}^*$ , the identification  $\text{Ker } i_n^* = \text{Hom}(H_n(C), G)$  is certainly natural, and the proof of Lemma 3.1 shows that  $\text{Ext}(H, G)$  depends naturally on  $H$ .

However, the splitting in the universal coefficient theorem is not natural since it depends on the choice of the projections  $p: C_n \rightarrow Z_n$ . An exercise at the end of the section gives a topological example showing that the splitting in fact cannot be natural.

The naturality property together with the five-lemma proves:

**Corollary 3.4.** *If a chain map between chain complexes of free abelian groups induces an isomorphism on homology groups, then it induces an isomorphism on cohomology groups with any coefficient group  $G$ .  $\square$*

One could attempt to generalize the algebraic machinery of the universal coefficient theorem by replacing abelian groups by modules over a chosen ring  $R$  and  $\text{Hom}$  by  $\text{Hom}_R$ , the  $R$ -module homomorphisms. The key fact about abelian groups that was needed was that subgroups of free abelian groups are free. Submodules of free  $R$ -modules are free if  $R$  is a principal ideal domain, so in this case the generalization is automatic. One obtains natural split short exact sequences

$$0 \rightarrow \text{Ext}_R(H_{n-1}(C), G) \rightarrow H^n(C; G) \xrightarrow{h} \text{Hom}_R(H_n(C), G) \rightarrow 0$$

where  $C$  is a chain complex of free  $R$ -modules with boundary maps  $R$ -module homomorphisms, and the coefficient group  $G$  is also an  $R$ -module. If  $R$  is a field, for example, then  $R$ -modules are always free and so the  $\text{Ext}_R$  term is always zero since we may choose free resolutions of the form  $0 \rightarrow F_0 \rightarrow H \rightarrow 0$ .

It is interesting to note that the proof of Lemma 3.1 on the uniqueness of free resolutions is valid for modules over an arbitrary ring  $R$ . Moreover, every  $R$ -module  $H$  has a free resolution, which can be constructed in the following way. Choose a set of generators for  $H$  as an  $R$ -module, and let  $F_0$  be a free  $R$ -module with basis in one-to-one correspondence with these generators. Thus we have a surjective homomorphism  $f_0: F_0 \rightarrow H$  sending the basis elements to the chosen generators. Now repeat the process with  $\text{Ker } f_0$  in place of  $H$ , constructing a homomorphism  $f_1: F_1 \rightarrow F_0$  sending a basis for a free  $R$ -module  $F_1$  onto generators for  $\text{Ker } f_0$ . And inductively, construct  $f_n: F_n \rightarrow F_{n-1}$  with image equal to  $\text{Ker } f_{n-1}$  by the same procedure.

By Lemma 3.1 the groups  $H^n(F; G)$  depend only on  $H$  and  $G$ , not on the free resolution  $F$ . The standard notation for  $H^n(F; G)$  is  $\text{Ext}_R^n(H, G)$ . For sufficiently complicated rings  $R$  the groups  $\text{Ext}_R^n(H, G)$  can be nonzero for  $n > 1$ . In certain more advanced topics in algebraic topology these  $\text{Ext}_R^n$  groups play an essential role.

A final remark about the definition of  $\text{Ext}_R^n(H, G)$ : By the Exercise stated earlier, exactness of  $F_1 \rightarrow F_0 \rightarrow H \rightarrow 0$  implies exactness of  $F_1^* \leftarrow F_0^* \leftarrow H^* \leftarrow 0$ . This means that  $H^0(F; G)$  as defined above is zero. Rather than having  $\text{Ext}_R^0(H, G)$  be automatically zero, it is better to define  $H^n(F; G)$  as the  $n^{\text{th}}$  homology group of the complex  $\cdots \leftarrow F_1^* \leftarrow F_0^* \leftarrow 0$  with the term  $H^*$  omitted. This can be viewed as defining the groups  $H^n(F; G)$  to be unreduced cohomology groups. With this slightly modified definition we have  $\text{Ext}_R^0(H, G) = H^0(F; G) = H^* = \text{Hom}_R(H, G)$  by the exactness of  $F_1^* \leftarrow F_0^* \leftarrow H^* \leftarrow 0$ . The real reason why unreduced Ext groups are better than reduced groups is perhaps to be found in certain exact sequences involving Ext and Hom derived in §3.F, which would not work with the Hom terms replaced by zeros.

## Cohomology of Spaces

Now we return to topology. Given a space  $X$  and an abelian group  $G$ , we define the group  $C^n(X; G)$  of **singular  $n$ -cochains with coefficients in  $G$**  to be the dual group  $\text{Hom}(C_n(X), G)$  of the singular chain group  $C_n(X)$ . Thus an  $n$ -cochain  $\varphi \in C^n(X; G)$  assigns to each singular  $n$ -simplex  $\sigma: \Delta^n \rightarrow X$  a value  $\varphi(\sigma) \in G$ . Since the singular  $n$ -simplices form a basis for  $C_n(X)$ , these values can be chosen arbitrarily, hence  $n$ -cochains are exactly equivalent to functions from singular  $n$ -simplices to  $G$ .

The **coboundary map**  $\delta: C^n(X; G) \rightarrow C^{n+1}(X; G)$  is the dual  $\partial^*$ , so for a cochain  $\varphi \in C^n(X; G)$ , its coboundary  $\delta\varphi$  is the composition  $C_{n+1}(X) \xrightarrow{\partial} C_n(X) \xrightarrow{\varphi} G$ . This means that for a singular  $(n+1)$ -simplex  $\sigma: \Delta^{n+1} \rightarrow X$  we have

$$\delta\varphi(\sigma) = \sum_i (-1)^i \varphi(\sigma| [v_0, \dots, \hat{v}_i, \dots, v_{n+1}])$$